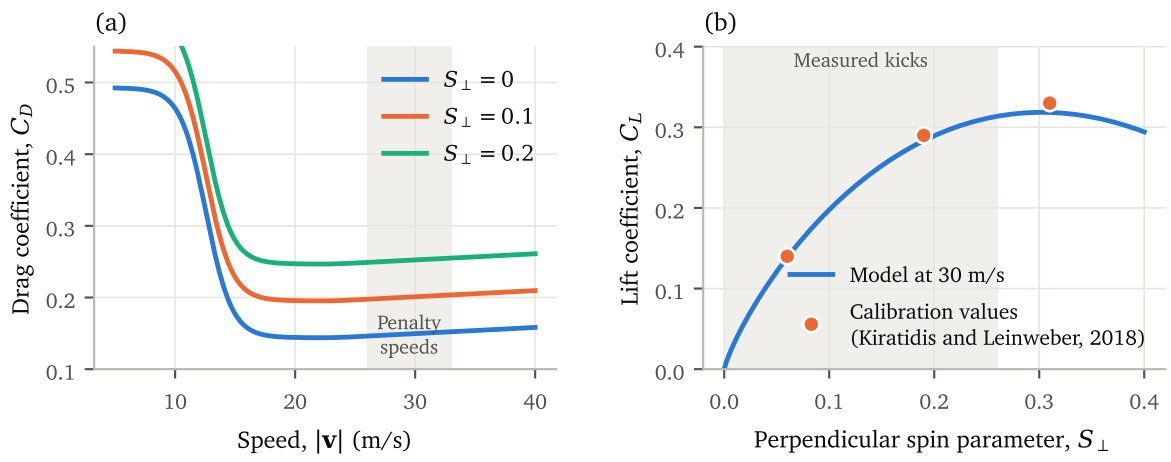
(a)
Drag coefficient, $C_D$
0.5
0.4
0.3
0.2
0.1
10
20
30
40
Speed, $|\mathbf{v}|$ (m/s)
$S_\perp = 0$
$S_\perp = 0.1$
$S_\perp = 0.2$
Penalty
speeds
(b)
Lift coefficient, $C_L$
0.4
0.3
0.2
0.1
0.0
0.0
0.1
0.2
0.3
0.4
Perpendicular spin parameter, $S_\perp$
Measured kicks
Model at 30 m/s
Calibration values
(Kiratidis and Leinweber, 2018)